

ALFWorld Tasks

→ Ongoing → Continue

→ Deviated → Correct

→ Completed → Transition

Task 1

Clean a plate and put on the diningtable.



Subgoal 1: Find and take the plate



Search for plate



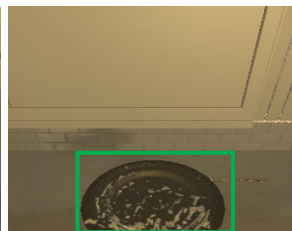
Take bowl 1

⊙ **Desired State** $\chi_{1,1}^d$
 $\langle \text{Inventory, plate, held} \rangle$
Execution State $s_{1,t}^e$
 Inventory: bowl 1 held
 ↓
Condition Relation
 Contradicted

Deviated
Corrective Reflection:
 I picked up the wrong object. I should put bowl 1 back and continue searching for plate.



Put back & Search plate



Take plate 1



Go to sinkbasin & clean

Subgoal 3: Place plate on diningtable



Go to diningtable & Place

Task 2

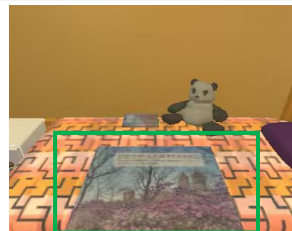
Look at book under the desk lamp.



Subgoal 1: Find and take the book



Go to sidetable 1



Find & Take book 2

⊙ **Desired State** $\chi_{1,1}^d$
 $\langle \text{Inventory, book, held} \rangle$
Execution State $s_{1,t}^e$
 Inventory: book 2 held
 ↓
Condition Relation
 Supported

Completed
Subgoal 1: Find and take the book
Memory 1: The robot is holding book 2 and observed desk lamp on sidetable 1

Subgoal 2: Find the desk lamp

⊙ **Desired State** $\chi_{2,1}^d$
 $\langle \text{Visible, desk lamp, true} \rangle$
Execution State $s_{2,t}^e$
 Inventory: book 2 held
 Target visible: book 1
 ↓
Condition Relation
 Unresolved

Ongoing
Think:
 Memory 1 indicates that the desk lamp is on the sidetable. I should go there to finish the subgoal.



Go to sidetable 1

Subgoal 3: Turn on the desk lamp



Use the desk lamp 1